

4. Decrypt the CT "BOOIQUYUBOSUOUTYRKAS". Encrypted using Rail fence with depth=3

→ Depth=3 Length=20

B		O		O		I		Q		
O	Y	U	B	O	S	N	U	U	T	
		R		K		A			S	

PT: BUY YOUR BOOKS IN AUGUST

5. Decrypt the cipher: "SEMSERTISINCMOO" encrypted using Rail Fence with depth=2.

→ Depth=2 Length=15

S	E	M	S	E	R	T	S	I
S	I	N	C	M	O	O		

PT: SSEIMN SCEMPTOI

Date:.....

MON	TUE	WED	THU	FRI	SAT	SUN
☐	☐	☐	☐	☐	☐	☐

Permutation

1. Encrypt message using key "53124"

WE ARE DISCOVERED.

→ 5 3 1 2 4

W E A R E

D I S C O

V E R E D

CT: ASR PCE EIE

EOD WDV

DES:

1. Using the Initial Permutation table, determine the output of the IP when the i/p is given in hexadecimal as: 0x002 0000 0000 0001

→

Binary: 0000 0000 0000 0010

Form 0000 0000 0000 0000

0000 0000 0000 0000

0000 0000 0000 0001

0000 0000 0000 0010 | 0000 0000 0000 0000 | 0000

0000 0000 0000 | 0000 0000 0000 0010

I/P bit 64 = 1 ; o/p position 25 = 7

I/P bit 15 = 7 ; o/p position 63 = 1

0000	0000	0000	0000		0000	0000	1000	0000
0000	0000	0000	0000		0000	0000	0000	0010

o/p in hexadecimal: 0x00 008 0000 0001

2. Given 4 bit binary PT(M) $\rightarrow$ 1101 & the key(K) $\rightarrow$ 1011. Calculate resulting CT(C) using bit by bit work.

PT $\rightarrow$	1	1	0	1
Key $\rightarrow$	1	0	1	1
CT $\rightarrow$	0	1	1	0
(XOR operation)				

3. The i/p to S-box 1 is 000000. What is the o/p?

$\rightarrow$ ~~14~~ 4 ~~13~~ 1 ~~2~~ 15 ~~11~~ 8 ~~3~~ 10 ~~6~~ 12

$\therefore$

0 0 0 0 0 0

1 1
C C

0th row & 0th column is
S-box is 14.

14 $\Rightarrow$ 1 1 1 0

O/p is = 1110